Week	Date	Topics	Readings	Assignments
1	M, Jul 27	**Course & Intelligent Systems Introduction** ● Syllabus, Grading and Methodology **Introduction to Artificial Intelligence.** ● History, Agents architecture and theory	Russell & Norvig, Ch. 1–2	
2	M, Aug 3	**Agent environments and architectures.** ● Intelligent agents. Rational agents **Ethical challenges and societal impact of AI** ● Ethical Design and AI case studies and discussion.	Russell & Norvig, Ch. 2–3 IEEE *Ethically Aligned Design*	Review for Midterm 1
3	M, Aug 10	**Midterm 1**		
4	**M, Aug 17**	**Holiday, No class**	—	
5	M, Aug 24	**Problem Solving by Search** ● BFS, DFS, Uniform Cost Search.	Russell & Norvig, Ch. 3	
6	M, Aug 31	**Problem Solving by Search** ● A* Search. Constraint Satisfaction Problems (CSP). Beam Search. Genetic Algorithms.	Russell & Norvig, Ch. 3–6	
7	M, Sep 7	**Adversarial Search** ● Game trees, Minimax algorithm, Alpha-Beta pruning	Russell & Norvig, Ch. 5	
8	M, Sep 14	**Local Search and Optimization** ● Hill climbing, Beam search, Simulated annealing		Review for Midterm 2
9	M, Sep 21	**Midterm 2**		
10	M, Sep 28	**Supervised Learning** ● Linear Models, KNN, Decision Trees ● Naïve Bayes, Neural Networks, SVM	Bishop, Ch. 3–4	
11	M, Oct 5	**Unsupervised Learning** ● K-Means and Hierarchical Clustering. **Reasoning Under Uncertainty** ● Bayesian Networks and Markov Models	Bishop, Ch. 9 Sutton & Barto, Ch. 1–6	
12	**M, Oct 12**	**Holiday, No class**	—	
13	M, Oct 19	**Model evaluation** ● Performance metrics, cross-validation ● ROC Curves and Overfitting	Goodfellow, Ch. 6; Bishop, Ch. 7	Review for Midterm 3
14	M, Oct 26	**Midterm 3**		
15	**M, Nov 2**	**Holiday, No class**	—	
16	M, Nov 9	Natural Language Processing ● Corpus Methods, Text Classification ● Sentiment Analysis, Automatic Summarization	Jurafsky & Martin, Ch. 1–8	Review for Project
17	**M, Nov 16**	**Holiday, No class**	—	—
18	M, Nov 23	**Course Project and Comprehensive Wrap-up**		

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